Huawei Access Node Optical Transport Failure Resolution SOP for KR-ACC-15

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2646

Description

Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access nodes occurs due to the degradation or physical damage of the optical components such as SFPs/XFPs, or due to issues with the optical fibers. This failure domain may also be triggered by incorrect optical signal power settings, which can be caused by human error during equipment installation or configuration. In the case of Huawei Access nodes, the SFP/XFP modules used for Optical Transport Network (OTN) transmission are prone to degradation due to the high-speed transmission rates and the harsh environmental conditions of the access network. Moreover, the use of single-mode fibers in the access network can increase the risk of fiber cuts, which can cause Optical Transport Failure. Therefore, it is essential to identify and rectify the underlying cause of the failure as soon as possible to ensure minimal downtime and maintain the overall network reliability.

Pre-requisite Checks

- Check: Verify that the KR Puram Access site is accessible and safe for dispatching remote hands., Description: Ensure that the site is secure and that the remote hands have the necessary clearance and equipment to perform the required tasks.

- Check: Verify the status of the redundant link KR-ACC-15., Description: Check if the redundant link is operational and can be used as a backup in case of further failures.

Potential Root Causes

Cause	Probability
SFP/XFP degradation or physical damage	High
Fiber cuts or physical damage	Medium
Incorrect optical signal power settings	Low

Diagnostic Commands & Expected Outputs

```
Command: display opt trans info, Expected Output: Optical transmission information for KR-ACC-15, Description: This command displays the optical transmission information for the KR-ACC-15 node, including the SFP/XFP module information, fiber status, and optical signal power levels. This will help identify if the SFP/XFP module is degraded or if there are any fiber issues.

Command: display opt trans sfp info, Expected Output: SFP/XFP module information for KR-ACC-15, Description: This command displays the SFP/XFP module information, including the module type, status, and optical signal power levels. This will help identify if the SFP/XFP
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module is degraded or if there are any issues with the optical signal
power settings.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace opt trans sfp module KR-ACC-15, Description: This command replaces the SFP/XFP module for the KR-ACC-15 node. Make sure to obtain the necessary replacement module and follow the correct installation procedures to ensure proper alignment and connection of the optical fibers.

Step 2: Step: Verify and adjust the optical signal power levels, Command: display opt trans sfp info, Description: This command displays the SFP/XFP module information, including the module type, status, and optical signal power levels. Use this information to verify that the optical signal power levels are within the recommended range. If necessary, adjust the optical signal power levels using the 'set opt trans sfp power' command.

Post-Check Verification

```
Command: display opt trans info, Expected Output: Optical transmission
information for KR-ACC-15 (resolved), Description: This command
verifies that the optical transmission information for the KR-ACC-15
node is within the normal range after the resolution steps.

Command: display opt trans sfp info, Expected Output: SFP/XFP module
information for KR-ACC-15 (resolved), Description: This command
verifies that the SFP/XFP module information is within the normal
range after the resolution steps.
```

Roll-back Procedures

- Step: Revert the SFP/XFP module replacement, Command: undo replace opt trans sfp module KR-ACC-15, Description: This command reverts the SFP/XFP module replacement for the KR-ACC-15 node. Make sure to follow the correct installation procedures to ensure proper alignment and connection of the optical fibers.